

Unit 7 Notes

Area, Surface Area, and Volume

How to use this file: Copy the notes carefully, then cover the worked examples and redo them without looking. For practice files, work first, then check the answer bank. Detailed solutions are included for Unit 3 through Unit 8 practice as requested.

Unit 7 Notes: Area, Surface Area, and Volume

Big goal

Measurement connects geometry to modeling. Area measures two-dimensional space, surface area measures the outside of a solid, and volume measures three-dimensional space. Units must match the dimension: square units for area and cubic units for volume.

1. Area formulas

Shape	Area formula
Rectangle	$A = bh$
Parallelogram	$A = bh$
Triangle	$A = \frac{1}{2}bh$
Trapezoid	$A = \frac{1}{2}(b_1 + b_2)h$
Rhombus or kite with diagonals	$A = \frac{1}{2}d_1d_2$
Regular polygon	$A = \frac{1}{2}aP$, where a is apothem and P is perimeter
Circle	$A = \pi r^2$
Sector	$A = \frac{\theta}{360^\circ}\pi r^2$ when θ is in degrees

2. Circumference and arc length

The circumference of a circle is $C = 2\pi r = \pi d$. An arc is a fraction of the circle:

$$\text{Arc length} = \frac{\theta}{360^\circ} \cdot 2\pi r.$$

3. Area on the coordinate plane

For horizontal or vertical bases, use base-height formulas. For any polygon with ordered vertices, use the shoelace formula:

$$A = \frac{1}{2} \left| \sum x_i y_{i+1} - \sum y_i x_{i+1} \right|.$$

Always list vertices in order around the polygon.

4. Surface area

Surface area is the total area of the exposed faces. A net is often the easiest way to see it.

- Rectangular prism: $SA = 2lw + 2lh + 2wh$.
- Cylinder: $SA = 2\pi r^2 + 2\pi rh$.
- Right prism: $SA = 2B + Ph$, where B is base area and P is base perimeter.
- Pyramid or cone: lateral area plus base area. For a right cone, $SA = \pi r^2 + \pi r\ell$, where ℓ is slant height.

5. Volume formulas

Solid	Volume formula
Prism	$V = Bh$
Cylinder	$V = \pi r^2 h$
Pyramid	$V = \frac{1}{3}Bh$
Cone	$V = \frac{1}{3}\pi r^2 h$
Sphere	$V = \frac{4}{3}\pi r^3$

6. Cavalieri's Principle

If two solids have the same height and equal cross-sectional area at every height, then they have the same volume. This explains why oblique prisms and cylinders have the same volume formulas as right prisms and cylinders.

7. Density and displacement

Density compares amount to size:

$$\text{density} = \frac{\text{mass}}{\text{volume}} \quad \text{or} \quad \frac{\text{people}}{\text{area}}.$$

Displacement is the volume of fluid moved by an object. If water in a cylinder rises, the displaced volume is

$$\pi r^2(\text{change in height}).$$

8. Modeling and design

Real objects can be modeled using geometric shapes. A tree trunk can be approximated by a cylinder, a roof by triangular prisms, and a storage tank by a cylinder plus hemispheres. In design problems, identify constraints, choose formulas, compute, and interpret units.

9. Worked examples

Example 1: Sector area

A sector has radius 12 and central angle 60° :

$$A = \frac{60}{360}\pi(12)^2 = 24\pi.$$

Example 2: Cylinder volume

A cylinder with radius 3 and height 10 has volume

$$V = \pi(3)^2(10) = 90\pi.$$

Example 3: Displacement

Water in a cylinder of radius 5 rises by 3. The displaced volume is

$$\pi(5)^2(3) = 75\pi.$$

Mastery checklist

You are ready for Unit 7 when you can choose area, surface area, and volume formulas, compute sector area and arc length, use coordinate area, apply density and displacement, and model real objects with geometric solids.